Software Requirements Specification

for

Inventory and Stock Management System (ISMS)

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1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) describes the functional and non-functional requirements for Version 1.0 of the Inventory and Stock Management System (ISMS). The ISMS is a web-based platform, supplemented by hands-free voice interaction on the warehouse floor, that allows Administrators, Managers, Warehouse Staff, and Sales Personnel to manage inventory, suppliers, purchase orders, and sales across physical and digital channels from a single system. This document is intended to be used by the project team responsible for the design, implementation, and verification of the system. Unless otherwise noted, all requirements specified here are of High or Medium priority and are targeted for Release 1.0.

1.2 Document Conventions

Functional requirements are identified using a dot-notation tag of the form Feature.Sub-feature.Detail (for example, Voice.Command.Interpret), following the convention used in this document's reference template. Each System Feature in Section 4 states its own priority (High, Medium, or Low); detailed requirements within a feature inherit that priority unless individually noted otherwise. Bold text is used for requirement tags and defined terms; italic text is used for stimulus/response labels.

1.3 Intended Audience and Reading Suggestions

This document is intended for the development team, quality assurance engineers, project mentors, and prospective stakeholders evaluating the proposed scope of the ISMS as a case study project. Readers seeking a high-level understanding should read Sections 1 and 2. Developers and testers should focus on Sections 3, 4, and 5, which contain the detailed, verifiable requirements. Section 4 is organized by system feature, so readers may jump directly to the feature relevant to their work.

1.4 Product Scope

The Inventory and Stock Management System is a unified platform that digitizes the end-to-end lifecycle of stock management - from supplier onboarding and purchase ordering through voice-enabled warehouse operations, omnichannel sales fulfillment, and predictive reordering. The product replaces fragmented manual processes (spreadsheets, paper stock registers, disconnected marketplace dashboards) with a single system shared by Administrators, Managers, Warehouse Staff, and Sales Personnel. A distinguishing capability of Release 1.0 is voice-enabled inventory management, which allows Warehouse Staff to perform CRUD operations hands-free, and AI-driven forecasting, which predicts demand and recommends reorder points. The goal of Release 1.0 is to deliver the seven core capabilities listed in Section 2.2, providing a solid, extensible foundation for future releases (e.g., native mobile apps, multi-language voice support, additional marketplace integrations).

1.5 References

- Wiegers, Karl. Software Requirements Specification templates and examples, Process Impact (reference materials used to structure this document).
- <Inventory and Stock Management System Vision and Scope Document> (to be prepared separately).
- <Organization UI/UX Style Guide>, <version>.
- <Marketplace Webhook API Documentation>, <marketplace provider names>.
- <Speech-to-Text and Text-to-Speech Engine API Documentation>, <provider name>.

2. Overall Description

2.1 Product Perspective

The Inventory and Stock Management System is a new, self-contained web application with a companion hands-free voice interface for warehouse operations. It is not a follow-on to an existing system. The ISMS integrates with four categories of external systems: (1) online Marketplaces and the organization's own storefront, via webhooks, for omnichannel order and stock synchronization, (2) a Speech-to-Text/Text-to-Speech engine for voice-enabled interaction and audio warehouse alerts, (3) an ML Forecasting Service for demand prediction and reorder-point calculation, and (4) an E-mail/SMS Gateway and Payment/Invoicing Gateway for notifications and digital sales processing. Figure 1 illustrates the major external entities and data flows for Release 1.0.

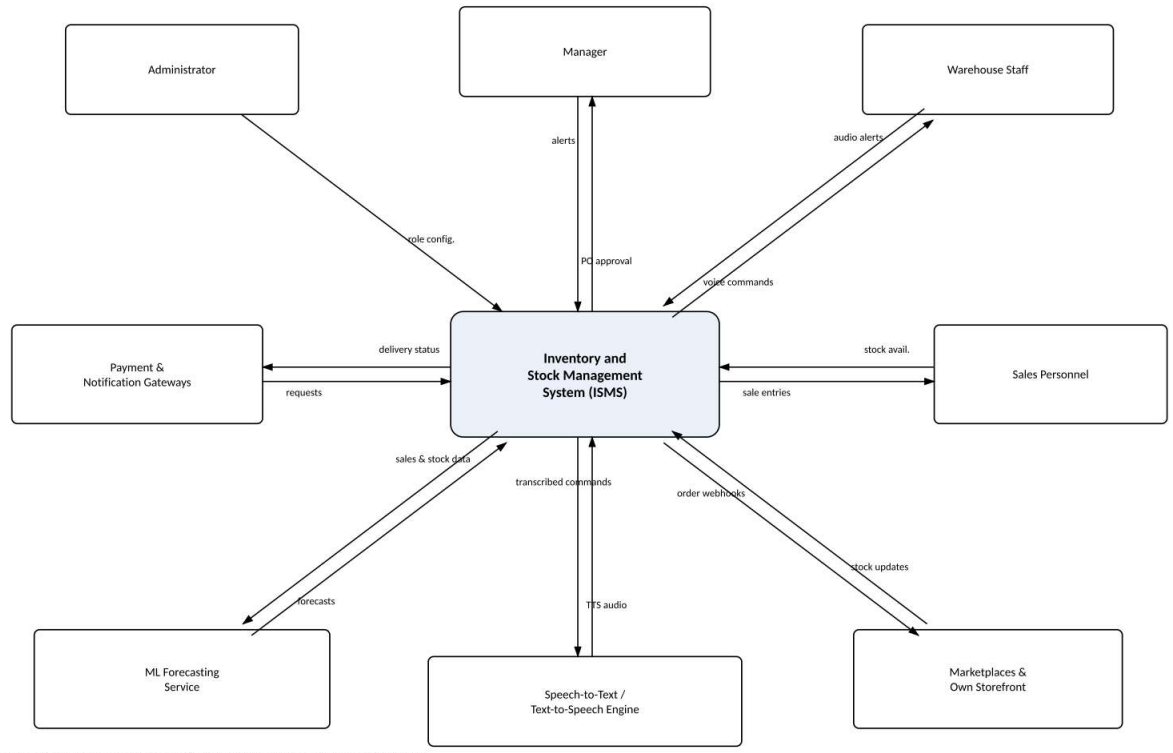


Figure 1. Context diagram for Release 1.0 of the Inventory and Stock Management System.

2.2 Product Functions

The major functions provided by Release 1.0 of the ISMS are summarized below; each is detailed as a System Feature in Section 4.

- Authentication & Role-Based Access Control - secure login and role-specific access for Administrators, Managers, Warehouse Staff, and Sales Personnel.
- Voice-Enabled Inventory & Omnichannel Sync - full CRUD operations on stock via standard UI or hands-free voice commands, synchronized across all sales channels in real time.

- Supplier Management - vendor profiles, contact details, performance history, and procurement records.
- Purchase Order Management - creation, approval, and tracking of inbound POs, with automatic stock updates on receipt.
- Sales Tracking & Order Management - logging of physical and digital sales, marketplace webhook processing, invoicing, and automated stock deduction.
- Predictive Analytics & AI Forecasting - dashboards for revenue and stock status, and ML-based prediction of seasonality, demand, and reorder points.
- Multimodal Notification System - real-time low-stock and shipping-delay alerts via text/e-mail and text-to-speech warehouse audio.

2.3 User Classes and Characteristics

User Class	Description
Administrator	Configures the system, manages user accounts and role permissions, oversees warehouses/stores, and has unrestricted access to all modules including analytics and audit logs. Technically proficient; infrequent but high-privilege use.
Manager	Oversees day-to-day inventory operations for one or more locations: approves purchase orders, reviews supplier performance, monitors dashboards and forecasts, and manages staff assignments. Frequent, daily use.
Warehouse Staff	Performs physical stock operations - receiving, put-away, picking, cycle counts - using the standard UI or hands-free voice commands on a handheld/wearable device. Largest operational user base; frequent use on the warehouse floor, often with hands occupied.
Sales Personnel	Logs in-store (physical) sales and manages customer-facing order queries at point-of-sale terminals; views real-time stock availability across channels before promising delivery dates. Frequent use during business hours.

2.4 Operating Environment

OE-1: The ISMS web application shall operate on current versions of Google Chrome, Mozilla Firefox, Microsoft Edge, and Safari.

OE-2: The ISMS shall be deployed on a cloud application server (Linux-based) with a horizontally scalable architecture to accommodate peak order-processing and voice-command loads.

OE-3: The voice-enabled warehouse interface shall operate on Android-based handheld scanners or wearable devices equipped with a microphone and speaker.

OE-4: The system shall permit access over the public Internet, secured by HTTPS/TLS, for all user classes.

2.5 Design and Implementation Constraints

CO-1: All personally identifiable information (staff and supplier contact details) shall be encrypted at rest and in transit.

CO-2: The system shall use a relational database engine to preserve referential integrity between products, stock records, suppliers, purchase orders, and sales orders.

CO-3: All externally facing APIs, including marketplace webhook endpoints, shall use JSON over HTTPS.

CO-4: The voice command module shall degrade gracefully to standard UI input when speech recognition confidence falls below a configurable threshold.

2.6 User Documentation

UD-1: The system shall provide contextual online help accessible from every major screen.

UD-2: The system shall provide a short first-time onboarding walkthrough for each of the Administrator, Manager, Warehouse Staff, and Sales Personnel roles, including a guided demonstration of supported voice commands.

2.7 Assumptions and Dependencies

AS-1: Warehouse Staff devices are assumed to have a functioning microphone and speaker and reasonable ambient noise conditions for voice recognition.

AS-2: Connected marketplaces provide a stable webhook mechanism for order events; the system's manual sales-entry function mitigates temporary webhook outages.

DE-1: The Predictive Analytics feature depends on a minimum volume of historical sales and stock data being available from the Sales Tracking and Purchase Order modules.

DE-2: The Multimodal Notification System depends on the continued availability of the Text-to-Speech engine and the E-mail/SMS gateway.

3. External Interface Requirements

3.1 User Interfaces

- UI-1: The ISMS shall provide role-specific dashboards for Administrators, Managers, Warehouse Staff, and Sales Personnel, each following a shared visual design system.
- UI-2: Every screen shall include a persistent help link and consistent primary navigation.
- UI-3: The Warehouse Staff interface shall provide a visible voice-command indicator showing listening, processing, and confirmation states.
- UI-4: Forms shall perform inline validation and display field-level error messages before submission.

3.2 Hardware Interfaces

- HI-1: The Voice-Enabled Inventory feature shall interface with the host device's microphone for command capture and speaker for audio confirmation and alerts.
- HI-2: The Sales Tracking feature shall interface with barcode scanners at point-of-sale terminals for physical sales entry.

3.3 Software Interfaces

Interface	Purpose
Marketplace Webhook Gateway	Receives real-time order notifications (created, paid, cancelled, returned) from connected online marketplaces and storefronts for omnichannel sync.
Speech-to-Text / Voice Command Engine	Converts spoken warehouse-staff commands (e.g., "add twenty units to bin A12") into structured inventory operations.
Text-to-Speech (TTS) Engine	Converts system-generated alerts and confirmations into spoken audio for hands-free warehouse broadcast.
ML Forecasting Service	Hosts the demand-forecasting and reorder-point models; accepts historical sales/stock data and returns predictions and recommended reorder quantities.
Payment / Invoicing Gateway	Processes payment capture for digital sales and generates itemized invoices for both physical and online orders.
E-mail / SMS Gateway	Delivers text-based low-stock, shipping-delay, and order alerts to registered users.

3.4 Communications Interfaces

- CI-1: The system shall send an e-mail or SMS alert to relevant Managers when a SKU's stock falls below its configured reorder point.
- CI-2: The system shall broadcast a text-to-speech audio alert over connected warehouse speakers for high-priority stock and shipping-delay events.
- CI-3: All communications containing payment or personal data shall be transmitted over encrypted channels.

4. System Features

This section organizes the functional requirements of the ISMS by system feature, corresponding to the seven core capability areas of Release 1.0.

4.1 Authentication & Role-Based Access Control

4.1.1 Description and Priority

Provides secure login for all users and restricts access to functions and data according to a user's assigned role (Administrator, Manager, Warehouse Staff, or Sales Personnel). Priority = High.

4.1.2 Stimulus/Response Sequences

Stimulus: *User submits login credentials.*

Response: *System verifies credentials and, if valid, grants access scoped to the user's assigned role.*

Stimulus: *Administrator assigns or changes a user's role.*

Response: *System updates the user's permitted functions and data access accordingly.*

Stimulus: *User attempts to access a function outside their role's permissions.*

Response: *System denies access and displays an appropriate message.*

4.1.3 Functional Requirements

Auth.Login: The system shall authenticate a user by username/employee ID and password before granting access to any function other than the login screen.

Auth.Login.Fail: If login credentials are invalid, the system shall display a generic error message and shall lock the account after 5 consecutive failed attempts within 15 minutes.

Auth.Password.Reset: The system shall allow a user to reset a forgotten password via a verified e-mail address.

Auth.RBAC.Roles: The system shall support at least four role types - Administrator, Manager, Warehouse Staff, and Sales Personnel - each with a distinct, configurable set of permissions.

Auth.RBAC.Assign: The system shall allow an Administrator to assign or change the role of any user account.

Auth.RBAC.Enforce: The system shall restrict access to every module and function strictly according to the permissions of the logged-in user's assigned role.

Auth.Session.Timeout: The system shall automatically log out a user after 30 minutes of inactivity.

Auth.Audit.Log: The system shall record a timestamped audit log entry for every login, logout, and role change.

4.2 Voice-Enabled Inventory & Omnichannel Sync

4.2.1 Description and Priority

Allows Warehouse Staff to perform full create, read, update, and delete (CRUD) operations on stock records using either the standard UI or hands-free voice commands, with every change synchronized in real time across all sales channels. Priority = High.

4.2.2 Stimulus/Response Sequences

Stimulus: *Warehouse Staff issues a spoken command to add or adjust stock at a bin location.*

Response: *System interprets the command, confirms the interpretation audibly, and updates the stock record upon verbal confirmation.*

Stimulus: *A stock record is created, updated, or deleted through any channel.*

Response: *System propagates the updated quantity to the physical store, the organization's own storefront, and all connected marketplaces.*

Stimulus: *Speech recognition confidence for a voice command is low.*

Response: *System asks the user to repeat the command or falls back to the standard UI.*

4.2.3 Functional Requirements

Inventory.CRUD.Create: The system shall allow an authorized user to create a new SKU with name, category, unit of measure, unit price, and reorder point.

Inventory.CRUD.Update: The system shall allow an authorized user to update the quantity on hand, location, or attributes of an existing SKU.

Inventory.CRUD.Delete: The system shall allow an authorized user to deactivate a SKU, provided it has zero quantity on hand across all locations.

Voice.Command.Capture: The system shall capture spoken input from a Warehouse Staff device and convert it to text using the Speech-to-Text engine.

Voice.Command.Interpret: The system shall parse the transcribed text into a structured inventory operation (action, SKU, quantity, location).

Voice.Command.Confirm: The system shall audibly repeat back the interpreted operation and require a verbal or tap confirmation before committing the change.

Voice.Command.LowConfidence: If speech recognition confidence falls below a configurable threshold, the system shall ask the user to repeat the command or switch to standard UI input.

Sync.Channel.Realtime: The system shall propagate any stock quantity change to all connected sales channels within 5 seconds.

Sync.Channel.Conflict: If two channels attempt to reserve the last available unit of a SKU simultaneously, the system shall grant the reservation to the first confirmed transaction and reject the other with an out-of-stock message.

Inventory.Search: The system shall allow a user to search and filter stock records by SKU, category, location, and quantity range.

4.3 Supplier Management

4.3.1 Description and Priority

Maintains vendor profiles, contact details, performance history, and procurement records to support informed sourcing decisions. Priority = Medium.

4.3.2 Stimulus/Response Sequences

Stimulus: *Manager adds a new supplier.*

Response: *System records the supplier's profile and contact details.*

Stimulus: *A purchase order involving a supplier is completed or delayed.*

Response: *System updates that supplier's performance history (on-time rate, quality flags).*

Stimulus: *Manager searches for a supplier by product category.*

Response: *System displays matching supplier profiles ranked by performance rating.*

4.3.3 Functional Requirements

Supplier.Create: The system shall allow a Manager or Administrator to create a supplier profile with company name, contact person, phone, e-mail, address, and supplied product categories.

Supplier.Edit: The system shall allow an authorized user to edit an existing supplier profile.

Supplier.Performance.Track: The system shall automatically update a supplier's on-time delivery rate and order-accuracy rate based on completed purchase orders.

Supplier.Performance.View: The system shall display each supplier's performance history, including past POs, delivery timeliness, and quality issue flags.

Supplier.Search: The system shall allow a user to search and filter suppliers by name, product category, and performance rating.

Supplier.Deactivate: The system shall allow an Administrator to deactivate a supplier, provided no purchase orders are currently open with that supplier.

4.4 Purchase Order (PO) Management

4.4.1 Description and Priority

Enables creation, approval, and tracking of inbound purchase orders, automatically increasing stock counts across all channels upon confirmed receipt. Priority = High.

4.4.2 Stimulus/Response Sequences

Stimulus: *Manager creates a purchase order for one or more SKUs from a supplier.*

Response: *System validates the order and routes it for approval.*

Stimulus: *Administrator approves a pending purchase order.*

Response: *System marks the PO as Approved and sends it to the supplier.*

Stimulus: *Warehouse Staff confirms receipt of a shipment against an approved PO.*

Response: *System increases stock quantities for the received items across all channels.*

4.4.3 Functional Requirements

PO.Create: The system shall allow a Manager to create a purchase order specifying supplier, one or more SKUs, quantities, and expected delivery date.

PO.Approve: The system shall require Administrator or Manager approval, per configured approval limits, before a purchase order is transmitted to the supplier.

PO.Approve.Reject: The system shall allow an approver to reject a purchase order with a reason, returning it to the creator for revision.

PO.Status.Track: The system shall track each purchase order through the statuses Draft, Pending Approval, Approved, Partially Received, Received, and Cancelled.

PO.Receive.Confirm: The system shall allow Warehouse Staff to confirm receipt of a shipment against an approved PO, specifying the quantity received per line item.

PO.Receive.Partial: If the quantity received is less than the quantity ordered, the system shall mark the PO as Partially Received and keep the outstanding quantity open.

PO.Receive.StockUpdate: Upon confirmed receipt, the system shall automatically increase the stock quantity on hand for each received SKU across all connected sales channels.

PO.Overdue.Flag: The system shall flag a purchase order as overdue if the expected delivery date has passed without full receipt.

4.5 Sales Tracking & Order Management

4.5.1 Description and Priority

Logs physical and digital sales, processes marketplace webhooks, generates invoices, and automatically deducts stock upon order confirmation. Priority = High.

4.5.2 Stimulus/Response Sequences

Stimulus: *Sales Personnel logs a physical, in-store sale.*

Response: *System deducts the sold quantity from stock and generates an invoice.*

Stimulus: *A connected marketplace sends an order webhook.*

Response: *System creates a corresponding sales order, reserves stock, and deducts quantity upon payment confirmation.*

Stimulus: *A customer requests a return or cancellation.*

Response: *System reverses the stock deduction and updates the order status accordingly.*

4.5.3 Functional Requirements

Sales.Physical.Log: The system shall allow Sales Personnel to log a physical sale by scanning or selecting SKUs and quantities at the point of sale.

Sales.Webhook.Receive: The system shall accept order-created, order-paid, order-cancelled, and order-returned webhook events from each connected marketplace.

Sales.Webhook.Validate: The system shall validate the authenticity of each incoming webhook using the marketplace's signature verification mechanism before processing it.

Sales.Order.Create: Upon a valid order event, the system shall create or update a sales order record with channel, line items, and status.

Sales.Stock.Deduct: The system shall automatically deduct the ordered quantity from stock on hand upon order confirmation, for both physical and digital sales.

Sales.Stock.Insufficient: If insufficient stock is available to fulfill an order, the system shall reject the order line and notify the responsible Manager.

Invoice.Generate: The system shall automatically generate a PDF invoice for every completed sale, itemizing products, quantities, prices, and taxes.

Sales.Return.Process: The system shall allow an authorized user to process a return or cancellation, reversing the stock deduction for the returned quantity.

Sales.Order.View: The system shall allow Sales Personnel and Managers to view and filter sales orders by channel, date range, and status.

4.6 Predictive Analytics & AI Forecasting

4.6.1 Description and Priority

Generates dashboards summarizing revenue and stock status, and uses machine learning models to predict seasonality, future demand, and optimal reorder points. Priority = Medium.

4.6.2 Stimulus/Response Sequences

Stimulus: *Manager opens the analytics dashboard.*

Response: *System displays current revenue, stock status, and trend charts across selected locations and date ranges.*

Stimulus: *The forecasting job runs on its scheduled interval.*

Response: *System generates updated demand forecasts and recommended reorder points for each SKU.*

Stimulus: *A recommended reorder point is generated for a SKU nearing its current reorder point.*

Response: *System suggests creating a purchase order pre-filled with the recommended quantity.*

4.6.3 Functional Requirements

Analytics.Dashboard.Revenue: The system shall display a dashboard summarizing revenue by channel, location, and time period.

Analytics.Dashboard.Stock: The system shall display a dashboard summarizing current stock status, including low-stock and overstock SKUs, across all locations.

Forecast.Demand.Generate: The system shall generate a demand forecast for each active SKU on a recurring schedule, using historical sales and seasonality data.

Forecast.Seasonality.Detect: The forecasting model shall account for detected seasonal patterns (e.g., recurring weekly or monthly demand cycles) in its predictions.

Forecast.Reorder.Recommend: The system shall calculate and display a recommended reorder point and reorder quantity for each SKU based on forecasted demand and supplier lead time.

Forecast.Reorder.Suggest: When a SKU's stock on hand approaches its recommended reorder point, the system shall suggest creating a pre-filled purchase order.

Forecast.Accuracy.Track: The system shall record forecast-versus-actual demand for each SKU to support ongoing model evaluation.

4.7 Multimodal Notification System

4.7.1 Description and Priority

Broadcasts real-time alerts for low stock and shipping delays through standard text/e-mail channels and automated text-to-speech warehouse audio. Priority = Medium.

4.7.2 Stimulus/Response Sequences

Stimulus: *A SKU's stock on hand falls below its reorder point.*

Response: *System sends a low-stock alert to the responsible Manager by e-mail/SMS and broadcasts an audio alert in the warehouse.*

Stimulus: *An approved purchase order becomes overdue.*

Response: *System sends a shipping-delay alert to the responsible Manager.*

Stimulus: *A user configures notification preferences.*

Response: *System sends future alerts only through the selected channels.*

4.7.3 Functional Requirements

Notify.LowStock.Trigger: The system shall generate a low-stock alert whenever a SKU's quantity on hand falls below its configured reorder point.

Notify.ShippingDelay.Trigger: The system shall generate a shipping-delay alert whenever a purchase order is flagged overdue per PO.Overdue.Flag.

Notify.Channel.EmailSMS: The system shall deliver text-based alerts via e-mail and/or SMS to the recipients configured for that alert type.

Notify.Channel.Audio: The system shall convert high-priority alerts to speech using the Text-to-Speech engine and broadcast them over connected warehouse audio devices.

Notify.Preferences.Configure: The system shall allow a user to configure which alert types they receive and through which channels (e-mail, SMS, audio).

Notify.Log: The system shall log every notification sent, including type, channel, recipient, and delivery status.

5. Other Nonfunctional Requirements

5.1 Performance Requirements

PE-1: The system shall support at least 100 concurrent Warehouse Staff issuing voice commands without response times exceeding 3 seconds per command.

PE-2: The system shall propagate a stock quantity change to all connected sales channels within 5 seconds of the triggering transaction.

PE-3: The system shall process an incoming marketplace webhook and update the corresponding sales order within 10 seconds of receipt.

PE-4: The scheduled demand-forecasting job shall complete for the full product catalog within 30 minutes.

5.2 Safety Requirements

SA-1: No safety-critical control functions (e.g., forklift or conveyor operation) are within the scope of this system; it provides inventory and order information only.

SA-2: Audio alerts broadcast in the warehouse shall not exceed configured volume limits to avoid interference with other safety announcements.

5.3 Security Requirements

SE-1: All network transactions involving payment or personally identifiable information shall be encrypted using TLS 1.2 or higher.

SE-2: Users shall authenticate before accessing any function other than the login screen.

SE-3: The system shall enforce role-based access control so that Administrator, Manager, Warehouse Staff, and Sales Personnel accounts can access only the functions and data permitted for their role.

SE-4: Voice command audio shall be transmitted to the Speech-to-Text engine over an encrypted channel and shall not be permanently stored beyond the interval needed for transcription.

SE-5: The system shall log all authentication, stock-changing, purchase-order, and sales transactions for audit purposes.

5.4 Software Quality Attributes

Availability-1: The ISMS shall be available 99.5% of the time, with priority uptime guaranteed during standard warehouse operating hours.

Usability-1: A first-time Warehouse Staff user shall be able to complete a basic voice-guided stock adjustment after a single onboarding walkthrough.

Scalability-1: The architecture shall support horizontal scaling to absorb order spikes following promotional sales events.

Robustness-1: If network connectivity is temporarily lost during a voice-command transaction, the system shall queue the command locally and sync it once connectivity is restored.

5.5 Business Rules

BR-1: Only a Manager or Administrator may create, approve, or reject a purchase order.

BR-2: A purchase order may not be approved if it would cause the supplier's outstanding order value to exceed a configurable credit limit.

BR-3: Stock on hand may never be reduced below zero; any sale or deduction exceeding available stock shall be rejected.

BR-4: A supplier's performance rating shall be recalculated only after a purchase order reaches Received or Cancelled status.

BR-5: Only an Administrator may deactivate a user account or a supplier profile.

6. Other Requirements

Internationalization: Release 1.0 targets English-language voice commands and UI text; the data model shall use UTF-8 encoding throughout to simplify future localization. Legal: The system shall present Terms of Service and a Privacy Policy for acceptance during account creation, and shall comply with applicable data-protection regulations for supplier and staff personal data. Reuse: Later releases may extend voice support to additional languages and add further marketplace integrations; this is out of scope for Release 1.0.

Appendix A: Glossary

Term	Definition
Bin / Location	A uniquely identified physical storage position within a warehouse or store to which stock keeping units are assigned.
Omnichannel Sync	The process of keeping stock counts consistent in real time across all sales channels (physical store, own website, and third-party marketplaces).
PO	Purchase Order - a formal request issued to a Supplier to procure a specified quantity of goods.
Reorder Point	The stock level at which a new purchase order should be triggered to avoid a stockout, calculated from lead time and forecasted demand.
SKU	Stock Keeping Unit - a unique identifier for a distinct product or product variant.
Supplier / Vendor	An external party from whom the organization procures inventory items.
Voice Command	A spoken instruction issued by Warehouse Staff and interpreted by the Speech-to-Text engine to perform an inventory CRUD operation.
Webhook	An automated, real-time HTTP callback sent by a marketplace to notify the system of a sales event.

Data Dictionary

The following data dictionary defines the key data elements referenced throughout this specification, listed in alphabetical order. Composite items are shown with their Page 15 of 18 component fields, marked “+”; enumerated values are shown in brackets; free-text or numeric fields are described between asterisks.

User

User =

user ID
name
email
role
hashed password
status

user ID = unique identifier assigned to each user account.

name = name of the user associated with the account.

email = e-mail address associated with the user account.

role = [Administrator | Manager | Warehouse Staff | Sales Personnel]

hashed password = securely stored representation of the user's password.

status = current status of the user account, such as active or inactive.

Product / SKU

product / SKU =

SKU ID
name
category
unit price
reorder point
unit of measure

SKU ID = unique identifier assigned to a distinct product or product variant.

name = name or description of the product.

category = category to which the product belongs.

unit price = selling price of one unit of the product.

reorder point = stock level at which the system recommends replenishment.

unit of measure = unit used to quantify the product, such as pieces, kilograms, or litres.

Stock Record

stock record =

SKU ID
warehouse/store ID
quantity on hand
quantity reserved
last updated

SKU ID = identifier of the product whose inventory is represented by the stock record.

warehouse/store ID = identifier of the warehouse or store location where the stock is held.

quantity on hand = current physical quantity of the SKU available at the specified location.

quantity reserved = quantity currently reserved for confirmed or pending sales orders.

last updated = date and time at which the stock record was most recently modified

Supplier

supplier =

supplier ID

name
contact details
rating
lead time
procurement history

supplier ID = unique identifier assigned to a supplier.

name = name of the supplier or vendor.

contact details = telephone number, e-mail address, address, and other contact information for the supplier.

rating = supplier performance rating based on delivery and order-accuracy history.

lead time = expected time between placing a purchase order and receiving the goods.

procurement history = record of previous purchase orders and supplier performance.

Purchase Order

purchase order =

PO ID
supplier ID
line items
status
order date
expected/received date

PO ID = unique identifier assigned to a purchase order.

supplier ID = identifier of the supplier to whom the purchase order is issued.

line items = list of products and quantities requested in the purchase order.

status = [Draft | Pending Approval | Approved | Partially Received | Received | Cancelled]

order date = date on which the purchase order was created.

expected/received date = expected delivery date or actual date on which the ordered goods were received.

The purchase-order status values above directly correspond to the lifecycle specified in your SRS.

Sales Order

sales order =

order ID
channel
customer information
line items
status
payment reference

order ID = unique identifier assigned to a sales order.

channel = sales channel through which the order was received, such as physical store, own storefront, or connected marketplace.

customer information = information associated with the customer placing the order.

line items = products and quantities included in the sales order.

status = current processing status of the sales order.

payment reference = reference identifying the payment transaction associated with the sales order.

The sales order entity supports both physical and digital sales, including marketplace webhook processing and stock deduction.

Forecast

forecast =

SKU ID
location ID
forecast period
predicted demand
recommended reorder quantity

SKU ID = identifier of the product for which demand is being predicted.

location ID = identifier of the warehouse or store location associated with the forecast.

forecast period = future period for which demand is predicted.

predicted demand = estimated quantity of the SKU expected to be required during the forecast period.

recommended reorder quantity = quantity recommended by the forecasting system for replenishing stock.

The forecasting requirements specify that forecasts use historical sales and seasonality information and produce recommended reorder points and reorder quantities.

Notification

notification =

notification ID
type
channel
recipient(s)
message
timestamp
status

notification ID = unique identifier assigned to a notification.

type = [Low Stock | Shipping Delay | Order Update | Other Alert]

channel = [E-mail | SMS | Audio]

recipient(s) = user or users who are intended to receive the notification.

message = content of the notification.

timestamp = date and time at which the notification was generated or sent.

status = delivery status of the notification.

The notification requirements specifically include low-stock and shipping-delay alerts and support e-mail, SMS, and audio delivery channels.

A.1 Data Relationships

The principal relationships between the entities are:

A User manages or operates on multiple Products/SKUs.
A Product/SKU can have multiple Stock Records across warehouses or stores.
A Supplier can be associated with multiple Purchase Orders.
A Purchase Order contains one or more products as line items.
A Sales Order contains one or more products as line items.
A Sales Order is associated with a payment reference.
A Product/SKU can have multiple Forecast records for different locations or forecast periods.
A User can receive multiple Notifications.
A Stock Record is associated with a particular product and warehouse/store location.

Appendix B: Analysis Models

B.1 Context Overview

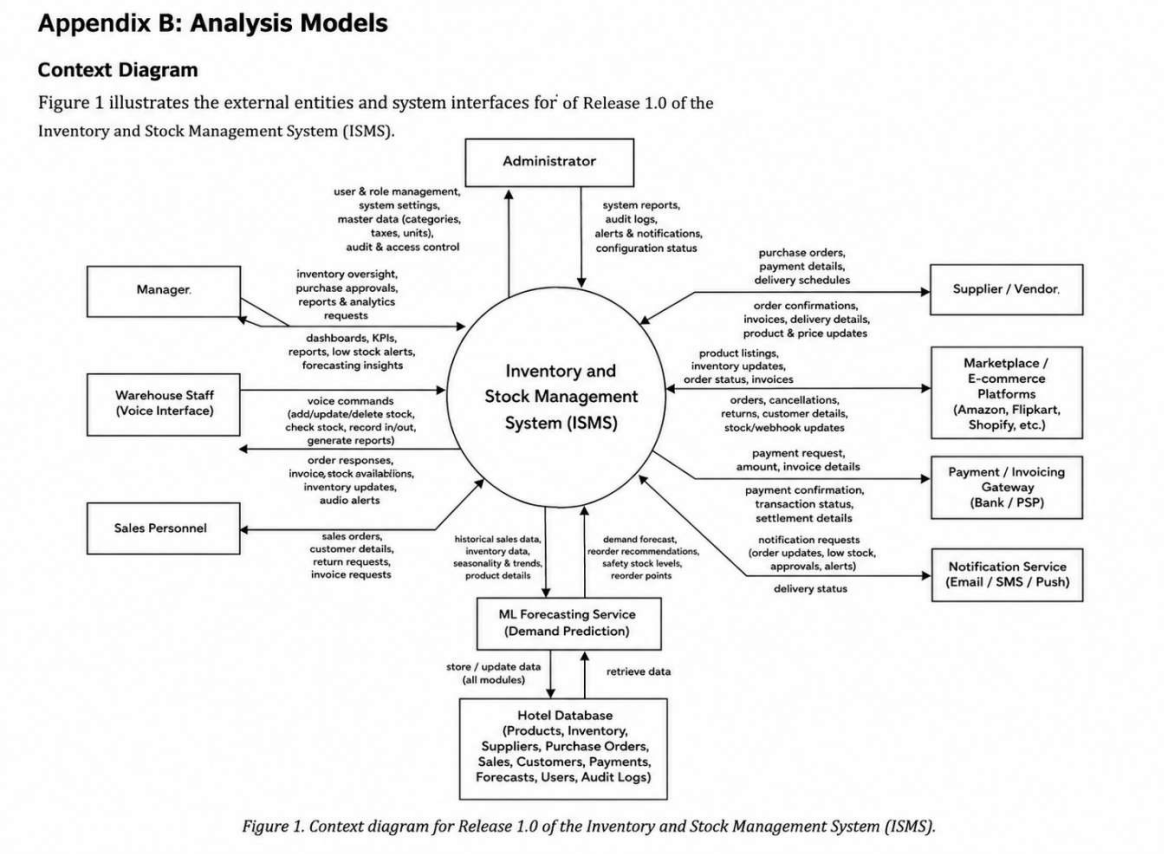
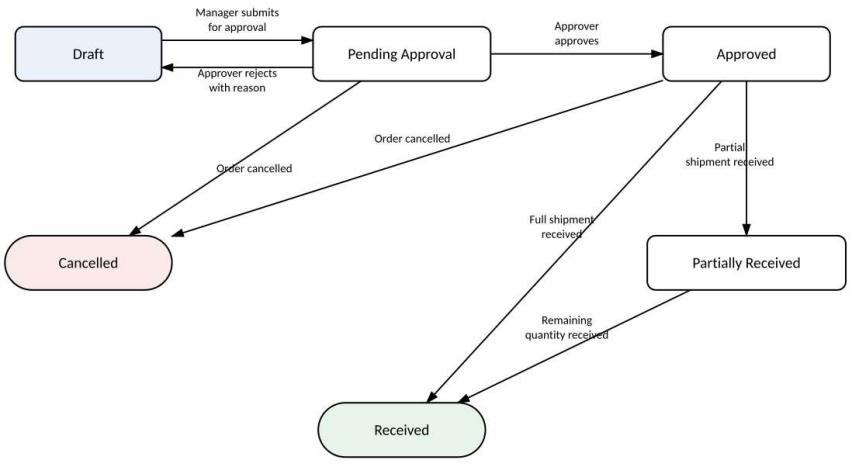


Figure 1. Context diagram for Release 1.0 of the Inventory and Stock Management System (ISMS).

B.2 Purchase Order Status - State Transitions

Figure 3 shows the state-transition diagram for purchase order status.



B.2 State Diagrams

B.2.1 Purchase Order State-Transition Diagram

Figure B.1 shows the lifecycle states of a purchase order and the events that cause it to move between states, referenced by Purchase Order management requirements in Section 4.4.

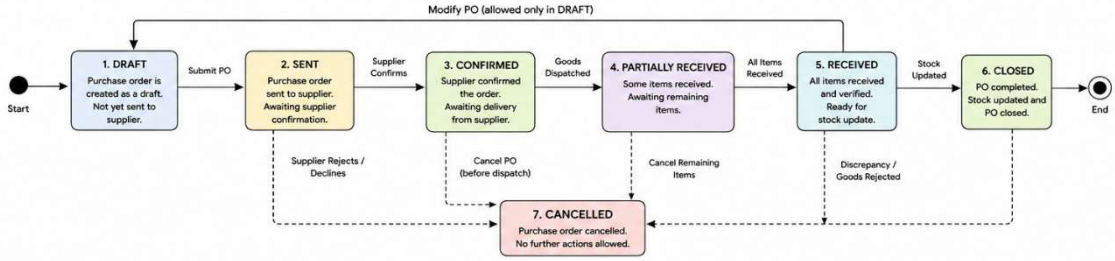


Figure B.1. State-transition diagram for purchase order status.

B.2.2 Sales Order / Fulfillment State-Transition Diagram

Figure B.2 shows the lifecycle states of a sales order (from placement to fulfillment) and the events that cause it to move between states, referenced by Sales Order management and Omnichannel Fulfillment requirements in Section 4.5.



Figure B.2. State-transition diagram for sales order / fulfillment status.

- Draft -> Pending Approval: Manager submits the purchase order for approval.
- Pending Approval -> Approved: Administrator or Manager approves the order within their approval limit.
- Pending Approval -> Draft: Approver rejects the order with a reason; it returns to the creator.
- Approved -> Partially Received: Warehouse Staff confirms receipt of less than the full ordered quantity.
- Partially Received / Approved -> Received: Warehouse Staff confirms receipt of all outstanding quantity.
- Approved / Pending Approval -> Cancelled: Manager or Administrator cancels the order before receipt.

B.3 Core Data Entities (Summary)

Figure 2 shows a partial data model for Release 1.0, followed by a summary of key attributes for each entity.

B.3 Partial Data Model

Figure B.3 shows a partial data model of the core entities described in Appendix A.

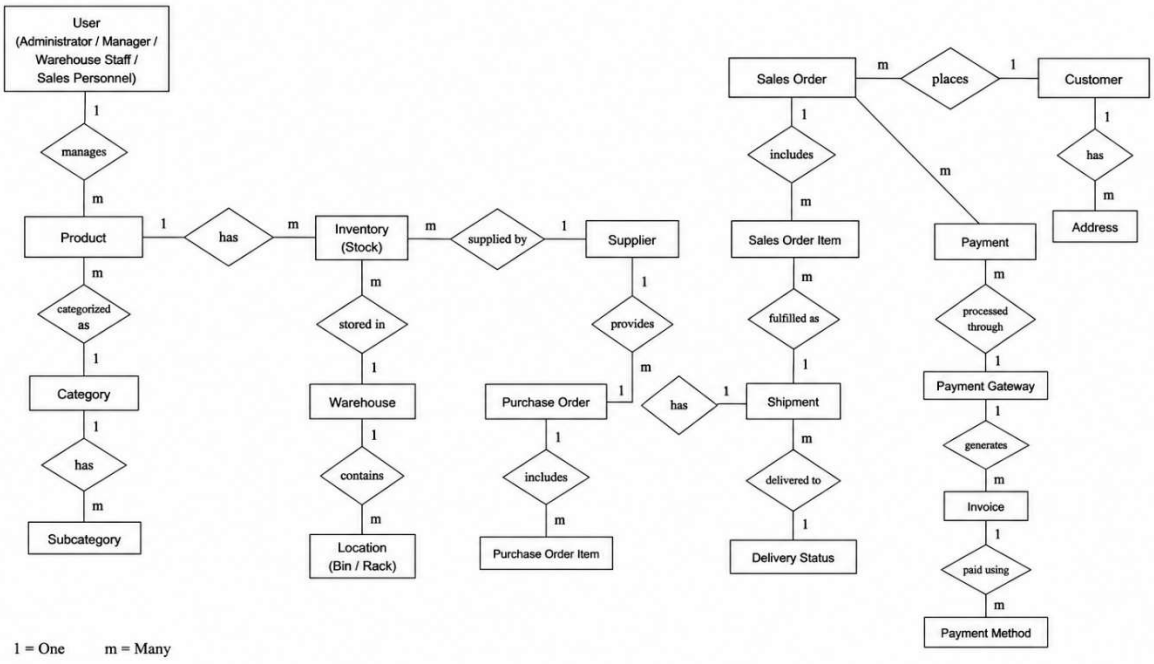


Figure B.3. Partial data model.

Entity	Key Attributes
User	user ID, name, email, role, hashed password, status
Product / SKU	SKU ID, name, category, unit price, reorder point, unit of measure
Stock Record	SKU ID, warehouse/store ID, quantity on hand, quantity reserved, last updated
Supplier	supplier ID, name, contact details, rating, lead time, procurement history
Purchase Order	PO ID, supplier ID, line items, status, order date, expected/received date
Sales Order	order ID, channel, customer info, line items, status, payment reference
Forecast	SKU ID, location ID, forecast period, predicted demand, recommended reorder qty
Notification	notification ID, type, channel, recipient(s), message, timestamp, status

Appendix C: To Be Determined List

#	TBD Item
TBD-1	Final list of supported voice command phrases and supported languages/accents for Release 1.0.
TBD-2	Specific marketplaces (e.g., Amazon, Flipkart) to be integrated in Release 1.0 versus later releases.
TBD-3	Choice of ML forecasting algorithm/library and minimum historical data required for reliable predictions.
TBD-4	Specific warehouse hardware (handheld scanners, wearable mic/speaker devices) to be certified for compatibility.
TBD-5	Data retention period for historical sales/stock data used in forecasting.